Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Menaker, Michael Ph.D.		
Project Title:	A Model of Circadian Disruption in the Space Environment		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	22904-4328	**Congressional District:**	5
Comments:	Deceased as of February 2021		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	07/01/2004	**End Date:**	06/30/2008
No. of Post Docs:	3	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Yoshikawa, Tomoko (University of Virginia)		
Grant/Contract No.:	NCC 9-58-HPF00406		
Performance Goal No.:			
Performance Goal Text:			
	(1) ORIGINAL AIMS OF PROJECT SPECIFIC AIM I: Using rhythms of gene expression reported by a luciferase transgene, we will identify the signals linking brain and peripheral oscillators. SPECIFIC AIM II: We will evaluate potential countermeasures by exposing our experimental animals to conditions that disrupt the normal circadian phase map and then determining if the potential countermeasures are able to restore it rapidly. SPECIFIC AIM III: Investigate the effects of chronic low dose methamphetamine in synchronizing disrupted rhythms and evaluate related drugs as potential countermeasures. (2) THE KEY FINDINGS OF THE PROJECT Using rat and mouse models, we have shown that the circadian system consists of hierarchically dominant oscillators in the central nervous system (CNS), and subordinate, but quasi-independent oscillators in the peripheral organs and tissues. There are at least 2 separate oscillators in the CNS. One, the suprachiasmatic nucleus (SCN) has been extensively studied. The other we have called the		

Task Description:	methamphetamine-sensitive circadian oscillator (MASCO) because it only manifests control over easily measured circadian end points when the subject is given chronic methamphetamine (or methylphenidate--Ritalin®). The properties of this second oscillator (MASCO) are not well researched, and we have focused some of our attention on it. MASCO can interact with the SCN oscillator, increasing the activity/rest ratio within a single cycle and lengthening the period of the activity rhythms. It is highly sensitive to the action of some drugs, particularly lithium, which, in combination with methamphetamine, lengthens its period from 26 to about 40 hours. We assume that MASCO is somewhere in the brain, but we don't know where. Our attempts to localize MASCO and elucidate its molecular mechanism are supported by a grant from the NIMH. The NSBRI support will be used to further characterize the effects of MASCO on circadian behavior and the physiology of peripheral circadian oscillators. Circadian oscillators in the periphery are kept in adaptive synchrony by signals from the dominant oscillators in the CNS. These signals are likely to be either neural, humeral or both, but at present, we know very little about them. We have begun to study the detailed relationships between central and peripheral oscillators by severing the neural connections between them. So far we have studied the effects of nerve section on ovaries and submaxillary glands with two quite different results. The phase of the circadian clock in the ovary is unaffected by denervation. This, coupled with experiments that show that ovarian phase can be reset by the pituitary hormones LH and FSH, suggest that phase control in the ovary is entirely hormonal. On the other hand, the phase of the oscillator in the submaxillary gland does not respond to food signals as long as its nerve supply is intact, but when the nerves are cut, the time of feeding becomes the only determinant of its phase. This suggests that it responds primarily to sympathetic neural input and secondarily to unknown circulating factors associated with food intake. (3) THE IMPACT OF THESE FINDINGS ON THE HYPOTHESES, OBJECTIVES AND SPECIFIC AIMS OF THE ORIGINAL PROPOSAL Our project is intended to provide some of the basic data that will be needed to minimize the effects of disrupting the circadian rhythms of astronauts and ground-based support personnel during their missions. Such disruption is an inevitable result of the work schedules that these men and women are required to follow. Disruption of circadian organization, which includes but is not limited to sleep deprivation, has repeatedly been shown to affect alertness, cognition, and performance on a variety of tasks. In order to minimize these effects, we have to understand the organization of the system that is being affected. (4) THE PROPOSED RESEARCH PLAN FOR THE COMING YEAR During the coming year we will: • Continue to study the behavioral and physiological responses of the circadian system to methamphetamine, methylphenidate, and lithium. • Extend our studies of neural and humeral coupling mechanisms in the circadian system to other peripheral organs. • Begin the study of potential countermeasures that will minimize circadian disruption by measuring their effects on the coupling between the CNS and the periphery.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There are several potential earth-based applications of our work that could generally improve human health. • Methylphenidate may be useful for temporarily stabilizing circadian phase in the face of disrupting environments. • Sympathetic nervous system agonists and antagonists of several kinds may be used to modify phase relationships among some peripheral circadian oscillators and thus improve the outcomes of chronic therapeutic treatments of cancer and other chronic conditions (i.e., high blood pressure and irregular heart beat). • The reproductive cycles--in particular their circadian components--of women engaged in activities with disruptive schedules may be regularized by carefully timed application of LH or FSH. • Our data suggest that shift work schedules in which cycles are repeatedly delayed will be less damaging to the health of workers than those in which these cycles are repeatedly advanced.
Task Progress:	• We have defined the phase shifting effects of preparation time on cultured peripheral circadian oscillators. This work has revealed unexpected complexity in the system [J Biol Rhythms 20(6): 500-512, 2005]. • We have described the methamphetamine oscillator in a mouse model and discovered the remarkable synergy between the effects of methamphetamine and lithium on free running period. The circadian effects of methamphetamine (or its analogue, Ritalin®) may be useful in stabilizing the circadian system in disruptive environments [J Biol Rhythms 21(3):185-194, 2006]. • We have shown that temporally restricted feeding alters the phase relationship between the circadian rhythms of healthy liver tissue and cancerous tissue from the same liver. This may be useful in designing chronotherapy regimens for treating cancer (International Journal of Cancer 118: 1623-1627, 2006). • We have shown that advancing phase shifts, but not delaying phase shifts, shorten the lives of aged mice, and that some, but not all, central and peripheral circadian oscillators shift with different trajectories in old vs. young rats (two manuscripts submitted). • We have made significant progress in defining the roles of neural and hormonal links in maintaining the integrity of the mammalian circadian system.
Bibliography Type:	Description: (Last Updated: 03/07/2011)
Articles in Peer-reviewed Journals	Davidson AJ, Yamazaki S, Arble DM, Menaker M, Block GD. "Resetting of central and peripheral circadian oscillators in aged rats." Neurobiol Aging. in press, June 2006. , Jun-2006
Articles in Peer-reviewed Journals	Davidson AJ, Straume M, Block GD, Menaker M. "Daily timed meals dissociate circadian rhythms in hepatoma and healthy host liver." Int J Cancer. 2006 Apr 1;118(7):1623-7. PMID: 16231323 , Apr-2006
Articles in Peer-reviewed Journals	Yoshikawa T, Yamazaki S, Menaker M. "Effects of preparation time on phase of cultured tissues reveal complexity of circadian organization." J Biol Rhythms. 2005 Dec;20(6):500-12. PMID: 16275769 , Dec-2005
Articles in Peer-reviewed Journals	Tataroglu O, Davidson AJ, Benvenuto LJ, Menaker M. "The methamphetamine-sensitive circadian oscillator (MASCO) in mice." J Biol Rhythms. 2006 Jun;21(3):185-94. PMID: 16731658 , Jun-2006